VLSI Task 1: Verilog RTL and Testbench

Power Control Top-Level Prototype — Xilinx Vivado

Use the first code block as a Design Source named **power_control_top.v** and the second as a Simulation Source named **tb_power_control_top.v**.

1. Verilog RTL Code

```
module power_control_top (
    input clk,
    input rst_n,
    input [7:0] voltage_sample,
    input [7:0] current_sample,
    input [7:0] duty_cmd,
    output reg pwm_out,
    output fault
);

    reg [7:0] counter;

    // Example fault thresholds
    assign fault = (voltage_sample > 8'd200) ||
                   (current_sample > 8'd150);

    // PWM generation
    always @(posedge clk or negedge rst_n) begin
        if (!rst_n) begin
            counter <= 8'd0;
            pwm_out <= 1'b0;
        end
        else begin
            counter <= counter + 1'b1;

            if (!fault && (counter < duty_cmd))
                pwm_out <= 1'b1;
            else
                pwm_out <= 1'b0;
        end
    end

endmodule
```

2. Verilog Testbench Code

```
`timescale 1ns / 1ps

module tb_power_control_top;

    reg clk = 0;
    reg rst_n = 0;
    reg [7:0] voltage_sample = 0;
    reg [7:0] current_sample = 0;
    reg [7:0] duty_cmd = 0;

    wire pwm_out;
    wire fault;

    power_control_top uut (
        .clk(clk),
        .rst_n(rst_n),
        .voltage_sample(voltage_sample),
        .current_sample(current_sample),
        .duty_cmd(duty_cmd),
        .pwm_out(pwm_out),
        .fault(fault)
    );

    always #5 clk = ~clk;

    initial begin
        // Reset
        #20 rst_n = 1;

        // Test 1: normal samples; no fault expected
        duty_cmd = 8'd128;
        voltage_sample = 8'd100;
        current_sample = 8'd50;
        #100;

        // Test 2: overvoltage; fault expected
        voltage_sample = 8'd220;
        current_sample = 8'd50;
        #50;

        // Test 3: overcurrent; fault expected
        voltage_sample = 8'd100;
        current_sample = 8'd160;
        #50;

        // Test 4: normal samples again
        current_sample = 8'd50;
        #50;

        $finish;
    end

endmodule
```

Expected behavior

Test 1: voltage=100, current=50 → fault=0. Test 2: voltage=220, current=50 → fault=1. Test 3: voltage=100, current=160 → fault=1. Test 4: voltage=100, current=50 → fault=0.

Notes

The fault thresholds (200 for voltage and 150 for current) are example values for simulation practice only. The samples are already digital; this does not implement a physical ADC. PWM is disabled when fault is asserted. The 8-bit counter wraps around naturally.